

Project Proposal: ConvoSched

1. Problem

Comic and anime conventions like Anime Expo publish massive schedules — hundreds of panels, screenings, and ceremonies across multiple days, rooms, and overlapping time slots. For Anime Expo 2026, the [official schedule](#) is easy to lose track of and offers no way to build a personal itinerary without manual copy-paste into a calendar app.

ConvoSched lets a user describe what they want in plain English — either "add the Welcome Ceremony to my calendar" (a known panel) or "cosplay gathering at field 5 on July 3rd 2pm" (a fan-created event outside the official schedule) — and have it parsed, matched against the real convention schedule when relevant, and added to a single personal planner with one confirmation step.

2. Intended Users

The primary user is **myself**, as an attendee planning my Anime Expo 2026 schedule.

More broadly, the tool could scale to other Anime Expo attendees building a personal itinerary, or to other convention organizers uploading their own schedules.

The current version assumes a single shared user/event list (no authentication); per-user authentication can be added in a later stage.

3. Cloud Platform and Services

The project is built entirely on **Google Cloud / Firebase**:

- **Firebase Hosting** — Serves the static frontend (plain HTML/CSS/JS) at convosched.web.app.
- **Cloud Functions for Firebase** — Three HTTP endpoints: [parseEvent](#) (natural-language command interpretation), [events](#) (CRUD on the personal calendar), and [catalog](#) (import/list/clear of the pre-parsed convention schedule).
- **Vertex AI — Gemini ([gemini-2.5-flash](#))** — Classifies a request as create/update/delete/clarify, extracts structured event fields (title, time, location, description) with JSON-schema-constrained output, and matches free-text requests against the convention catalog.

- **Cloud Firestore (Native mode)** — Two collections: `events` (personal calendar) and `catalogEvents` (437 pre-parsed Anime Expo 2026 panels, imported once via a Gemini-based parsing pipeline).

All AI calls happen server-side via Cloud Functions using the function's service account — no API keys live in the frontend. A single `firebase deploy` updates hosting, functions, and Firestore rules/indexes together.

4. Type of AI Assistant

ConvoSched is a **task-oriented conversational assistant for planner management**, combining two AI capabilities:

1. **Natural-language command interpretation.** Every user input is sent to Gemini with structured-output (JSON schema) prompting, which classifies the intent (`create` / `update` / `delete` / `clarify`) and extracts the relevant fields. For updates and deletes, Gemini is given a summary of the user's existing events to resolve references like "move my lunch with Sarah." Ambiguous or incomplete requests fall back to a `clarification` response with a follow-up question, rather than guessing.
2. **Catalog-grounded retrieval/matching.** A one-time pipeline uses Gemini to parse the raw Anime Expo 2026 schedule text into 437 structured catalog entries stored in Firestore. At request time, a compact catalog summary is given to Gemini alongside the user's request, so a phrase like "Maid Cafe on July 3 at noon" is matched to the correct entry — merging its description, room, and exact times into the proposed event.

In both cases, the assistant never writes to the database directly: it returns a proposed event for the user to review, edit if needed, and confirm — keeping the human in control.